The Plant Doctor · Sample test results

Run on 27 Oct 2026 in Chromium using Demo Coggy (simulated sensors), field physics off, custom field time. Each reading was held for 5 checks (5 seconds). No real ESP32 was used, so this tests the app logic, not the wiring. ✓ means the decision matches the Plant Doctor's advice: water only dry soil when it is not raining, and switch lights on only when it is dark.

1. Setup

Brain	How it was set up
Automation	Rule 1: If time is 6:00 AM to 6:30 AM, then Pump ON. Rule 2: If it is dark, then Lights ON.
AI	Trained on the built-in example diary: 5 days, 24 moments (Day 3 rainy, Day 4 wet soil, Day 5 cloudy). It matches the diary 100% for the pump and 100% for the lights. What it learned: Pump ON when soil is dry and it is not raining; Pump OFF when soil is wet; Pump OFF when soil is dry and it is raining; Lights ON when it is dark; Lights OFF when it is bright.

2. Sample readings and decisions

#	Situation	Time	DHT-11	Soil	Rain	LDR	Doctor says	Auto pump	Auto lights	AI pump	AI lights	AI reason (pump)
1	**Morning watering time, dry soil**	6:05 AM	24°C · 70%	Dry	No	Bright	Pump ON, lights OFF	ON ✓	OFF ✓	ON ✓	OFF ✓	Soil is dry and it is not raining. Seen 8 times, 100% sure
2	**Watering time, rain starts (soil still dry)**	6:10 AM	23°C · 88%	Dry	Yes	Bright	Pump OFF, lights OFF	ON ✗	OFF ✓	OFF ✓	OFF ✓	Soil is dry and it is raining. Seen 2 times, 100% sure
3	**Watering time, raining, soil soaked**	6:15 AM	23°C · 92%	Wet	Yes	Bright	Pump OFF, lights OFF	ON ✗	OFF ✓	OFF ✓	OFF ✓	Soil is wet. Seen 14 times, 100% sure
4	**Mid-morning, wet soil**	9:00 AM	27°C · 66%	Wet	No	Bright	Pump OFF, lights OFF	OFF ✓	OFF ✓	OFF ✓	OFF ✓	Soil is wet. Seen 14 times, 100% sure
5	**Hot noon, dry soil**	12:30 PM	36°C · 32%	Dry	No	Bright	Pump ON, lights OFF	OFF ✗	OFF ✓	ON ✓	OFF ✓	Soil is dry and it is not raining. Seen 8 times, 100% sure
6	**Cloudy afternoon, dark, wet soil**	2:00 PM	25°C · 78%	Wet	No	Dark	Pump OFF, lights ON	OFF ✓	ON ✓	OFF ✓	ON ✓	Soil is wet. Seen 14 times, 100% sure
7	**Evening, dark, dry soil**	7:30 PM	26°C · 60%	Dry	No	Dark	Pump ON, lights ON	OFF ✗	ON ✓	ON ✓	ON ✓	Soil is dry and it is not raining. Seen 8 times, 100% sure
8	**Night, dark, wet soil**	10:00 PM	23°C · 70%	Wet	No	Dark	Pump OFF, lights ON	OFF ✓	ON ✓	OFF ✓	ON ✓	Soil is wet. Seen 14 times, 100% sure

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3. Live test totals (40 checks each)

Measure	Automation	AI
Plant Doctor score (live)	75%	100%
Water used	0.75 L	0.75 L
Checks wasting water	10 (25%)	0 (0%)
Thirsty checks (dry soil, no water)	10 (25%)	0 (0%)
Lights on in daylight	0	0
Plant health at the end	71%	100%
Times it switched	3	9

4. Challenge round (Compare page)

Situation	Time	Automation pump · lights	AI pump · lights	Doctor says
Watering time, but it is raining	6:05 AM	ON ✗ · OFF ✓	OFF ✓ · OFF ✓	OFF · OFF
Watering time, soil already wet	6:05 AM	ON ✗ · OFF ✓	OFF ✓ · OFF ✓	OFF · OFF
Hot afternoon, dry soil	2:30 PM	OFF ✗ · OFF ✓	ON ✓ · OFF ✓	ON · OFF
Dark storm clouds at noon	12:00 PM	OFF ✓ · ON ✓	OFF ✓ · ON ✓	OFF · ON
Night, soil has dried out	9:30 PM	OFF ✗ · ON ✓	ON ✓ · ON ✓	ON · ON

Overall Plant Doctor score on the Compare page (13 situations): Automation 69%, AI 100%.

5. What the results show

• The app worked end to end with no errors: rules, live decisions, pump and light commands, AI training and the Compare page.

• Automation did exactly what its rules said. At 6:10 and 6:15 AM it kept the pump ON while it was raining and the soil was soaked, because the time rule has no rain check. At hot noon and in the evening it left dry soil unwatered because those times are outside the 6:00 to 6:30 AM window.

• AI turned the pump OFF in the rain and when the soil was wet, and watered dry soil at noon and in the evening. It did this because the example diary showed those choices, not because anyone wrote a rule for them.

• Both brains handled the lights correctly, because the Automation rule "lights on when dark" already matched what the AI learned.

• The AI only looks this good because the training diary was sensible. If students farm carelessly during the 5 days, the AI copies their mistakes. That is a good talking point for the class.

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